

Tutorial 7: Data Integration and Linkage

Question 1

Consider the following three parties related to the *Olympics* information system involving swimming events:

- Local Organisation Committee (LOC) for an Olympiad
- International Olympic Committee (IOC)
- FINA, the international sporting body governing swimming

Assume the following tables are owned and managed by the LOC, IOC, and FINA:

- LOC: *Result*(EventID, CompID, Position, Time)
- IOC: *Competitor*(ID, Country, Name)
OlympicRecord(EventID, CompID, Olympiad, Time)
- FINA: *Athlete*(ID, Country, Name)
WorldRecord(EventID, AthID, Year, Time)

Suppose that all competitors participating the Game organised by the LOC are registered with the IOC already (i.e., LOC.CompID is a foreign key to IOC.ID), and that each competitor is also already registered by the relevant international sporting body, in this case FINA.

- Identify possible semantic heterogeneity when integrating these three independently developed databases into *GoldMedallist*, and discuss possible solutions.
- Now the LOC wants to integrate these three databases into the following table that shows all swimming event Gold medallists in the Game:

GoldMedallist(CompID,EventID,Time,OlympicRecord,WorldRecord)

Use SQL to construct *GoldMedallist*. (Hint: using views)

- Assume the ABC Television wants to create the following table to show all the swimming records set at the Game organised by the LOC:

NewRecord(EventID, CompID, Record, Time)

where *Record* is either "World" or "Olympic". Show an SQL query computing the table *NewRecord*.

Question 2

Answer the following questions about string similarity measures.

- (a) What is *Edit distance/Levenshtein Metric*?
- (b) What is *Jaccard coefficient*?
- (c) Calculate the *similarity* between the following strings using *Edit distance* and *Jaccard coefficient* (3-gram) respectively.
 - "University Queensland"
 - "Queensland University"
- (d) Which one is better for measuring the similarity between the above two strings, Edit distance or Jaccard coefficient, and why?

Question 3

Answer the following questions:

- (a) What problem is solved by Cosine Similarity using the TF/IDF measure?
- (b) If the numerical numbers of 1,000 and 999 are regarded as similar data items, how do we compute this kind of similarity?

Question 4

See the following two tables' schema:

DRIVER

Driver-License#	First-Name	Middle-Name	Family-Name	DoB	Gender	Address	Home-Phone-Number	Expiry-Date
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TAXPAYER

Tax-File #	First-Name	Surname	Resident Status	DoB	Sex	Post-Code	eMail	Contact#
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We want to create a mapping table namely, PERSON based on these two given tables:

PERSON

Tax-File #	Driver-License#
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How do we do it?

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